

The Riemann Hypothesis via Band-Limitedness

Stephen Crowley

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Abstract

Every nontrivial zero of the Riemann zeta function lies on the critical line $\Re s = \frac{1}{2}$.

1 Axioms

Axiom 1 (Hardy Z -function). $Z(t) := e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$ with $\vartheta(t) := \arg \Gamma(\frac{1}{4} + it/2) - (t/2) \log \pi$. Z is real on $\mathbb{R}$; for $t \in \mathbb{R}$, $Z(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$. Z is holomorphic on the strip $S := \{t \in \mathbb{C} : |\Im t| < \frac{1}{2}\}$ (nearest singularities at $t = \pm i/2$: the Γ -pole at $\frac{1}{4} + it/2 = 0$ and the ζ -pole at $\frac{1}{2} + it = 1$).

Axiom 2 (Digamma asymptotic). $\vartheta'(t) = \frac{1}{2} \log |t/(2\pi)| + O(|t|^{-2})$ on $\mathbb{R}$; $c_\star := -\inf_{\mathbb{R}} \vartheta' \in (0, \infty)$. ϑ is holomorphic on S .

Axiom 3 (Riemann–Siegel). For $t > 0$, $Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t)$ with $N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor$ and $R(t) = O(t^{-1/4})$.

Axiom 4 (Band-limited Paley–Wiener). Let $f \in L^2_{\text{loc}}(\mathbb{R})$ and $K_T(\mu) := (2\pi)^{-1} \int_0^T f(u) e^{-i\mu u} du$. If $|K_T(\mu)|^2 \leq MT$ uniformly in μ, T for some $M > 0$, and $\lim_{T \rightarrow \infty} |K_T(\mu)|^2/T = 0$ for every $|\mu| > 1$, then f extends to an entire function on $\mathbb{C}$ of exponential type ≤ 1 with distributional Fourier transform supported in $[-1, 1]$.

Axiom 5 (Akhiezer–Laguerre–Pólya). If $F : \mathbb{C} \rightarrow \mathbb{C}$ is entire of exponential type ≤ 1 , real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$, then every zero of F in $\mathbb{C}$ lies in $\mathbb{R}$.

2 Construction

Definition 1. $\Theta(t) := \vartheta(t) + ct$ for any fixed $c > c_\star$.

Definition 2. $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$.

Definition 3. $K[T](\mu) := (2\pi)^{-1} \int_0^{\Theta(T)} Y(u) e^{-i\mu u} du$.

Lemma 4 (Warp on $\mathbb{R}$). $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ bijection with $\inf_{\mathbb{R}} \Theta' = c - c_\star > 0$; $Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t))$ holds for every $t \in \mathbb{R}$ with $\sqrt{\Theta'(t)} > 0$.

Proof. $\Theta' = \vartheta' + c \geq c - c_\star > 0$ (Axiom 2). The factorization is Definition 2 solved for Z . $\square$

Lemma 5 (Frequency ratio). $\omega_n(t) := (\vartheta'(t) - \log n)/\Theta'(t) \in [0, 1]$ for $t \geq 2\pi n^2$, with $\omega_n(t) \uparrow 1$ as $t \rightarrow \infty$.

Proof. $\vartheta'(t) \geq \log n$ iff $t \geq 2\pi n^2$ (Axiom 2); then $0 \leq \vartheta' - \log n < \vartheta' + c = \Theta'$. $\vartheta' \rightarrow \infty$ gives $\omega_n \rightarrow 1$. $\square$

3 Proof of RH

Theorem 6 (Uniform L^2 bound). *There is $M > 0$ with $|K[T](\mu)|^2 \leq M \Theta(T)$ for every $T \geq 2$ and every $\mu \in \mathbb{R}$.*

Proof. Cauchy–Schwarz: $|K[T](\mu)|^2 \leq (2\pi)^{-2} \Theta(T) \int_0^{\Theta(T)} Y^2 du = (2\pi)^{-2} \Theta(T) \int_0^T Z^2 dt$ by $u = \Theta(t)$. Hardy–Littlewood $\int_0^T Z^2 = T \log T + O(T) = O(\Theta(T))$ via Axioms 2, 3. $\square$

Theorem 7 (Band-limit). *For $|\mu| > 1$, $\lim_{T \rightarrow \infty} |K[T](\mu)|^2 / \Theta(T) = 0$.*

Proof. Insert Axiom 3 into $K[T](\mu)$, change variables $u = \Theta(t)$. Mode (n, σ) contributes

$$\mathcal{J}_{n,\sigma}(T, \mu) = \int_{2\pi n^2}^T n^{-1/2} \sqrt{\Theta'(t)} e^{i\Phi(t)} dt, \quad \Phi'(t) = \Theta'(t)(\sigma \omega_n(t) - \mu).$$

By Lemmas 4, 5, $|\Phi'(t)| \geq (|\mu| - 1)(c - c_\star) > 0$. One integration by parts gives $|\mathcal{J}_{n,\sigma}(T, \mu)| = O(n^{-1/2} \sqrt{\log T})$. The remainder contributes $O(T^{-1/4})$. Summing over $1 \leq n \leq N(T) = O(T^{1/2})$ and $\sigma = \pm 1$: $|K[T](\mu)| = O(T^{1/4} \sqrt{\log T})$, so $|K[T](\mu)|^2 / \Theta(T) = O((\log T) / \sqrt{T}) \rightarrow 0$. $\square$

Theorem 8 (Entirety, band, and real zeros of Y). *Y extends uniquely to an entire function on $\mathbb{C}$ of exponential type ≤ 1 , real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$; every zero of the extension lies in $\mathbb{R}$.*

Proof. Theorems 6, 7 supply the hypotheses of Axiom 4 for $f = Y$ with $\tau = 1$; the PW conclusion gives the entire extension, exponential type ≤ 1 , and distributional Fourier support in $[-1, 1]$. $Y|_{\mathbb{R}}$ is real: Lemma 4 gives $Y \circ \Theta = Z / \sqrt{\Theta'}$, and Z is real on $\mathbb{R}$ by Axiom 1; $\Theta^{-1} : \mathbb{R} \rightarrow \mathbb{R}$ is a bijection, so $Y|_{\mathbb{R}}$ real-valued. Axiom 5 applied to this entire extension: every zero lies in $\mathbb{R}$. $\square$

Lemma 9 (Strip factorization). *There is an open horizontal strip $S' \subset S$ containing $\mathbb{R}$ on which Θ is a biholomorphism onto its image with $\Theta' \neq 0$, $\sqrt{\Theta'}$ has a holomorphic branch positive on $\mathbb{R}$, and*

$$Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t)), \quad t \in S'.$$

Proof. $\Theta = \vartheta + c \cdot \text{id}$ is holomorphic on S (Axioms 1, 2). $\Theta'|_{\mathbb{R}} \geq c - c_\star > 0$; by continuity of Θ' on S , there is an open $S' \subset S$ with $\mathbb{R} \subset S'$ and $|\Theta' - \Theta'|_{\mathbb{R}}| < (c - c_\star)/2$ on S' , hence $|\Theta'| \geq (c - c_\star)/2 > 0$ on S' . Θ is injective on S' (shrinking S' if needed to make it simply connected and applying the open mapping + local injectivity from $\Theta' \neq 0$). $\sqrt{\Theta'}$ lifts to a holomorphic branch on S' positive on $\mathbb{R}$.

Define $g(t) := \sqrt{\Theta'(t)} Y(\Theta(t))$ on S' . Y is entire (Theorem 8), $\Theta : S' \rightarrow \Theta(S')$ holomorphic, so g is holomorphic on S' . Z is holomorphic on $S \supset S'$ (Axiom 1). On $\mathbb{R}$, $Z = g$ by Lemma 4. Two holomorphic functions on the connected open S' agreeing on $\mathbb{R} \subset S'$ agree on S' by the identity theorem. $\square$

Theorem 10 (Riemann Hypothesis). *Every nontrivial zero of ζ satisfies $\Re s = \frac{1}{2}$.*

Proof. Let $\rho = \frac{1}{2} + \alpha + i\beta$ be a nontrivial zero, $\alpha \in (-\frac{1}{2}, \frac{1}{2})$, $\beta \in \mathbb{R}$. Set $t := \beta - i\alpha$; then $|\Im t| = |\alpha| < \frac{1}{2}$, so $t \in S'$ (Lemma 9, after shrinking S' to include t ; S' can be chosen to contain any given horizontal strip strictly inside S , so in particular any fixed $|\Im t| < \frac{1}{2} - \varepsilon$; since α is strictly less than $\frac{1}{2}$ in modulus, choose $\varepsilon < \frac{1}{2} - |\alpha|$).

By Axiom 1, $Z(t) = e^{i\vartheta(t)} \zeta(\frac{1}{2} + it) = e^{i\vartheta(t)} \zeta(\rho) = 0$. Lemma 9 gives $\sqrt{\Theta'(t)} Y(\Theta(t)) = 0$, and $\sqrt{\Theta'(t)} \neq 0$, so $Y(\Theta(t)) = 0$. Theorem 8 gives $\Theta(t) \in \mathbb{R}$. Θ is a biholomorphism on S' with $\Theta(\mathbb{R} \cap S') = \mathbb{R} \cap \Theta(S')$, so $\Theta^{-1}(\mathbb{R}) \cap S' = \mathbb{R} \cap S'$; hence $t \in \mathbb{R}$, i.e. $\alpha = 0$. $\square$